

Chip War Endgame 2030

Who Controls Advanced Semiconductors Controls the 21st Century

Blue Lotus Research Intelligence | August 2026

Classification: Open Intelligence

Executive Summary

The global semiconductor industry is the economic and military battleground upon which the 21st century's dominant power relationship will be determined. The United States' decision to impose sweeping export controls on advanced semiconductor technology access to China — beginning in October 2022 and progressively expanded through 2025 — represents the most consequential industrial policy decision since the US-Japan Semiconductor Agreement of 1986. Unlike that earlier trade dispute, the current US-China semiconductor conflict is not about market access or intellectual property theft; it is about whether China will be permitted to develop the domestic capability to manufacture the advanced chips upon which artificial intelligence, hypersonic weapons, autonomous systems, and quantum computing depend.

As of August 2026, the chip war's first phase — establishing the export control framework — is complete. The second phase — determining whether China can break through the controls through indigenous technology development — is underway with preliminary results: China has demonstrated remarkable resilience and ingenuity (SMIC's 7nm workaround, YMTC's 232-layer NAND before control impact, Huawei's Mate 60 Pro with domestic chips), but faces fundamental physical limits (no EUV, no sub-5nm at commercial yields) that cannot be resolved by capital investment or policy will alone. The third phase — which is just beginning — is whether the US coalition (Japan, Netherlands, and potentially the EU and Korea) can sustain the export control framework against intense diplomatic pressure from China and internal commercial pressure from US semiconductor companies losing billions in Chinese revenue.

Part I — The Weapons of the Chip War

Export Controls: BIS's Expanding Toolkit

The US Commerce Department's Bureau of Industry and Security (BIS) has constructed a semiconductor export control architecture of remarkable complexity and ambition. The key instruments:

Entity List: Approximately 600+ Chinese entities as of August 2026 require US government licence (routinely denied for advanced semiconductor purposes) to receive US-origin goods, software, and technology. Huawei (2019), SMIC (2020), CXMT, Naura Technology, YMTC, and hundreds of others are listed.

Foreign Direct Product Rule (FDPR): The FDPR extends US controls to foreign-made products that use US-origin technology, software, or equipment in their production — even if the product itself contains no US-origin components. This is the mechanism that restricts TSMC (Taiwanese company, using US-origin EDA software and equipment) from manufacturing advanced chips for Huawei entities. The FDPR extension to the advanced computing chips context (October 2022 rule) is arguably the most

consequential single policy tool: it globally restricts advanced chip supply to China regardless of the chip's physical manufacturing location.

Advanced Computing Rule: Specific performance thresholds (total processing performance, memory bandwidth, interconnect bandwidth) above which chips cannot be exported to China without licence. NVIDIA's A100 and H100 GPUs — the training accelerators used for frontier AI model development — exceeded these thresholds. NVIDIA subsequently designed "China-compliant" variants (A800, H800) that stayed below thresholds; BIS updated the rules in October 2023 to restrict these as well, closing the loophole.

ASML's EUV: The Single Most Consequential Chokepoint

ASML Holding NV — headquartered in Eindhoven, Netherlands — is the world's sole manufacturer of Extreme Ultraviolet (EUV) lithography machines, the equipment required to manufacture chips below approximately 5nm with economically viable yields. Each EUV machine costs approximately \$150-200M (High-NA EUV machines, the next generation, cost approximately \$380M), is assembled from 100,000+ components sourced from approximately 5,000 suppliers in 40+ countries, and requires approximately 10 years of accumulated knowledge to operate at production yields.

ASML stopped shipping EUV machines to China in 2019 (before the US-led export control framework) at the request of the Dutch government, which declined to renew ASML's export licence for China. The 2023 Dutch export control expansion further restricted ASML's DUV (Deep Ultraviolet, previous generation) immersion lithography systems — which China had been using as an EUV workaround — from export to China below a specified wavelength/NA threshold.

The EUV chokepoint's significance cannot be overstated. Semiconductor chip manufacturing requires hundreds of process steps; EUV is used for approximately 15-20 critical patterning steps in advanced logic production. Without EUV, manufacturers must use multiple DUV passes (multi-patterning) to achieve equivalent patterning resolution — increasing process steps by approximately 3-4×, reducing yield, and increasing per-chip cost to a level that makes commercial competition with EUV-manufactured chips economically challenging. SMIC's demonstrated 7nm workaround is technically impressive but economically impractical at scale: the yield loss and additional process steps make SMIC's 7nm chips uncompetitive with TSMC's TSMC-N7 at equivalent specifications.

Part II — China's Capability: The 30-35% Reality

What China Has Achieved

China's domestic semiconductor industry, despite significant disadvantages, has achieved more than most Western analysts predicted in 2018-2019. As of August 2026:

- DRAM: CXMT (ChangXin Memory Technologies) has begun mass production of DDR5 DRAM at 19nm equivalent — comparable to industry-leading nodes of 2-3 years ago. CXMT is not at the Samsung/SK Hynix frontier but is competitive for applications that don't require the highest performance or HBM configurations.
- NAND: YMTC achieved 232-layer NAND Flash in 2023 before export controls froze its equipment supply. Current YMTC production is at 192-232 layers; without new advanced equipment, further scaling is constrained.
- Logic: SMIC's 7nm workaround (as demonstrated in the Huawei Kirin 9000s) is technically feasible at low volume. Commercial-scale 7nm production without EUV faces yield and cost challenges that limit its applications to government-priority programmes (military, HPC, strategic AI) rather than mass consumer chips.

- Equipment: NAURA Technology Group, Advanced Micro-Fabrication Equipment (AMEC), and other Chinese equipment companies have made significant progress in etch, deposition, and inspection tools. Chinese equipment's market share in China fabs has grown from approximately 10% (2019) to approximately 25-30% (2025), predominantly in mature node applications.
 - Overall self-sufficiency: China produces approximately 30-35% of the semiconductors it consumes (by value) domestically — against a stated target of 70% that has been repeatedly deferred. At mature nodes (28nm and above), China's domestic supply is growing; at advanced nodes (7nm and below), China remains overwhelmingly import-dependent from Taiwan, Korea, and other non-restricted sources.
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Part III — The US Coalition: Can It Hold?

Commercial Pressure on the Export Control Framework

The US semiconductor export control framework's biggest vulnerability is the commercial pressure it creates on US companies that derive significant revenue from China. NVIDIA's China AI chip revenue (before controls) was approximately \$4B annually; Lam Research, Applied Materials, and KLA together lost approximately \$6-8B in annual China equipment revenue due to export control restrictions. These companies are publicly-listed, have shareholders who expect financial performance, and employ engineers who develop the technologies that are being restricted — creating internal tensions between compliance and commercial interest.

The semiconductor industry's lobbying effort on export controls is the most sophisticated and best-funded in the technology sector's history. SIA (Semiconductor Industry Association) has consistently argued that overly broad export controls harm US competitiveness more than they impede China — by denying US companies revenue that funds R&D, reducing the talent attraction that competitiveness requires, and pushing China toward accelerated domestic development. The tension between national security (restrict as much as possible) and commercial/competitiveness (restrict as little as possible consistent with security) is embedded in the policy framework and will not be resolved.

Allies: Japan, Netherlands, and the Weak Links

The export control framework's effectiveness requires allied participation — the Wassenaar Arrangement cannot restrict items that non-member countries continue to export. The critical allies are Japan (TEL, Nikon, Advantest equipment) and the Netherlands (ASML). Both have implemented export controls aligned with the US framework, but with slightly different scope and exemption structures.

The framework's weakest element is the absence of formal Korean export control expansion to match Japan's. Korean companies — Samsung, SK Hynix, Wonik IPS (semiconductor equipment), and chemical suppliers — have not implemented the same equipment restrictions that Japan imposed in 2023. This gap creates a potential workaround: Chinese fabs can potentially source certain equipment from Korean suppliers that would be restricted from US/Japanese suppliers. The US has pressured Korea to align its export controls, but Korea's commercial exposure to China (including semiconductor fab operations there) creates political resistance to the level of control expansion that Japan accepted.

Part IV — The 2030 Endgame Scenarios

Scenario A: Export Controls Hold, China Hits a Ceiling (Probability: 40%)

The US-Japan-Netherlands trilateral framework holds without significant erosion; Dutch restrictions on advanced DUV systems prevent China from improving beyond 5nm at commercial yields. China's semiconductor industry achieves approximately 40% overall self-sufficiency by 2030 (up from 30-35% in

2026), primarily through mature node expansion. China's AI development is constrained — not stopped — by the inability to access frontier chips at the volumes required for the most compute-intensive frontier model training. China's military AI development prioritises efficiency (doing more with less compute) and develops application-specific chip designs optimised for specific military missions. The technology gap between US/allied AI and Chinese AI widens over the decade, providing the US and its allies a compounding advantage in AI-enabled military systems.

Scenario B: China Breaks the EUV Barrier (Probability: 20%)

China's domestic EUV development programme — estimated at approximately \$3-5B+ in government investment — achieves a functioning EUV source by 2028-2030. (Feasibility is not impossible: EUV was developed from first principles by ASML over 30 years; China is not starting from zero, has significant academic physics capability, and is using economic espionage and talent acquisition to accelerate.) A Chinese EUV equivalent — even if significantly less productive than ASML's NXE:3400 or High-NA systems — would break the chokepoint and enable Chinese progression to sub-5nm at commercial yields within 3-5 years of EUV capability. This scenario represents the chip war's decisive Chinese victory.

Scenario C: Coalition Fracture, Export Control Erosion (Probability: 25%)

US domestic political pressure (from semiconductor companies, from China-accommodating factions within both parties) erodes the export control framework. Key exemptions are granted; the FDPR is narrowed; allied export controls are not strengthened to match US intention. China accesses sufficient advanced equipment to close the manufacturing gap to 7nm at commercial scale by 2028. Advanced AI chip supply restriction relaxes — China's hyperscalers (Alibaba Cloud, Tencent Cloud, Baidu) access adequate compute for frontier model training. The technology gap narrows; the US loses its AI-enabled military systems advantage.

Scenario D: Conflict Resolution, New Framework (Probability: 15%)

US-China diplomatic engagement produces a technology trade framework that provides China some semiconductor access in exchange for Chinese commitments on Taiwan, cyber, and AI safety. This scenario seems implausible given current political dynamics but has historical precedent (Nixon to China, Reagan-Gorbachev arms control). It would represent a negotiated endgame rather than a competitive one.

Part V — Why Semiconductors Determine Strategic Power

The Military Connection

The US export control framework's national security rationale rests on a premise that requires explicit statement: advanced semiconductors are the enabling technology for every next-generation military system. The AI-enabled autonomous systems (drones, missile guidance, ISR processing), hypersonic maneuvering vehicles (requiring real-time aerodynamic calculations), directed-energy weapons (LIDAR, laser rangefinders), and cyber operations infrastructure all depend on chips at the frontier of compute performance. A nation that cannot manufacture advanced chips domestically — or cannot access them from allied suppliers — is constrained in its ability to develop and deploy these military systems at the speed and scale required for 21st-century conflict.

The 2030 endgame of the chip war will determine whether the US-allied technology lead in AI-enabled military systems narrows or widens. Every year China's chip manufacturing is constrained at 7nm and above, the US compounds its AI training and inference advantage. Every year China successfully develops domestic alternatives, it narrows the gap. The mathematics of technological compounding — where the leading side's advantage in AI capability enables better AI research, which enables more

advanced AI, in a recursive loop — means that the early years of the chip war are the most consequential.

Conclusion: The Century's Defining Industrial Competition

The semiconductor chip war is not a trade dispute, a technology competition, or a geopolitical rivalry. It is all three simultaneously, and its outcome will shape the military balance, economic structure, and political power distribution of the 21st century more profoundly than any other ongoing competition.

Northeast Asia — Taiwan (TSMC), Korea (Samsung/SK Hynix), and Japan (materials/equipment) — is the geographic centre of this competition: the producer of the contested capabilities, the location of the chokepoints both sides are trying to control or access, and the potential flashpoint if military conflict is used to resolve what economic and technological competition cannot.

The 2030 endgame is not predetermined. The US-allied coalition that controls the semiconductor supply chain has advantages that compound with time — but requires sustained political will, commercial sacrifice, and diplomatic coherence among partners with different commercial interests. China has shown more technological resilience than initially expected — but faces physical limits that cannot be overcome by ambition or capital alone.

The chips are set. The game is played for keeps.

Blue Lotus Research Intelligence | Northeast Asia Cross-Regional Series | Chip War Endgame 2030 | August 2026

All data from BIS, SIA, ASML, CSIS, Semianalysis, TechInsights, and cited news sources.